SMART INDIA HACKATHON 2026

TITLE PAGE

Problem Statement ID –SIH26025

Problem Statement Title- Development of an AI enabled

Low Cost Real Time Mine Subsidence Monitoring, Prediction and

Early Warning System for

Underground Coal Mines in India

Theme- SMART AUTOMATION

PS Category- Hardware

Team ID- 169894

Team Name (Registered on portal):TICH TITANS

IDEA TITLE

Tich Titans

SMART INDIA HACKATHON 2026

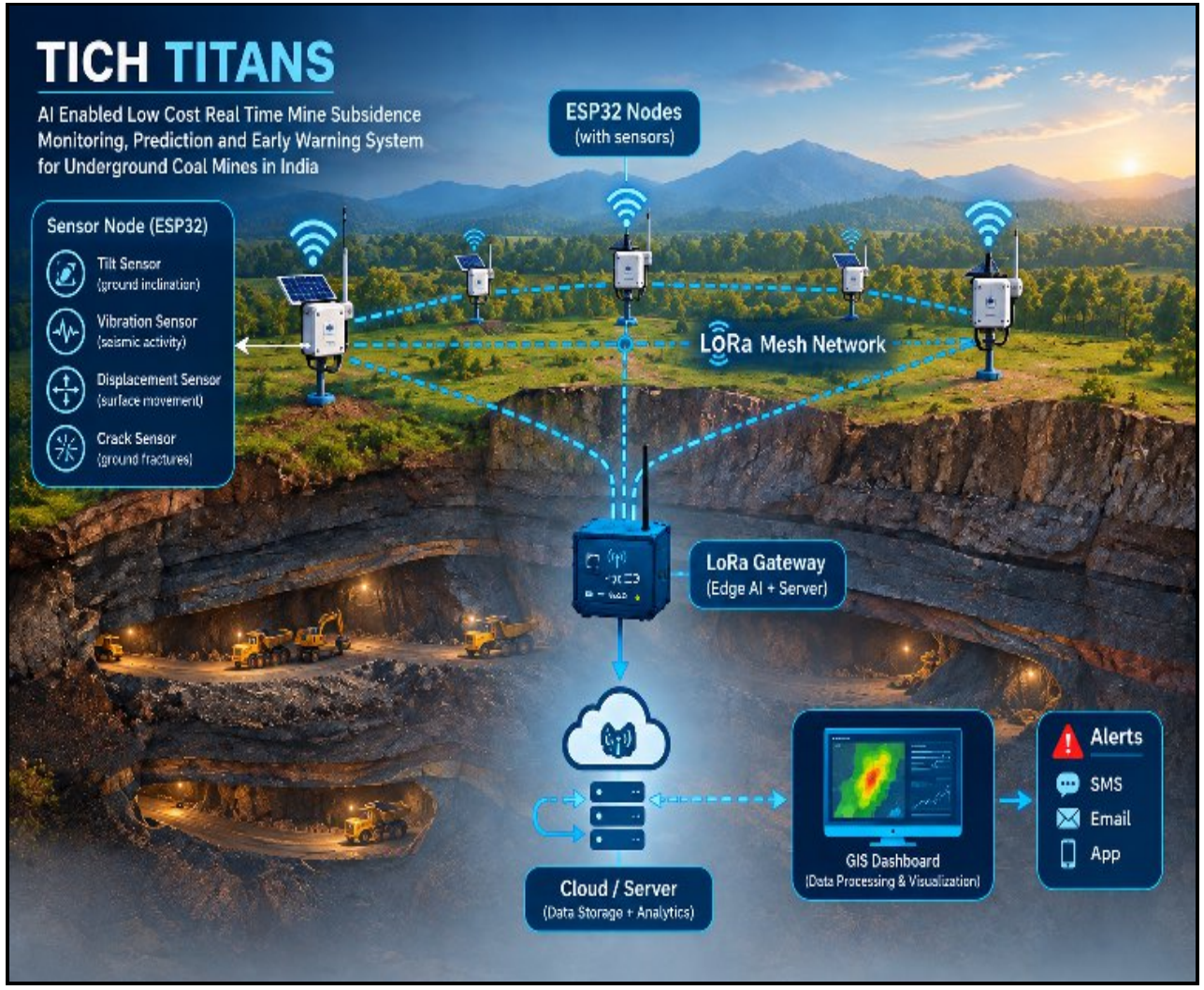
Idea/Solution paragraph:
Implementation of a low-cost wireless surface mesh sensor network above underground mine panels, using AI/ML to detect early ground deformation and issue automated subsidence warnings before critical failure.

ESP32 sensor nodes with tilt, vibration, displacement, and crack sensors continuously sense micro ground movement.

LoRa mesh networking relays data node-to-node to a central gateway, working even where cellular coverage is absent.

Edge AI (TinyML) on nodes flags anomalies locally; a **Raspberry Pi gateway ML model** predicts subsidence zones and severity.

GIS dashboard + automated alerts deliver real-time risk maps and SMS/email/app warnings to operators and communities.



Tich Titans

TECHNICAL APPROACH

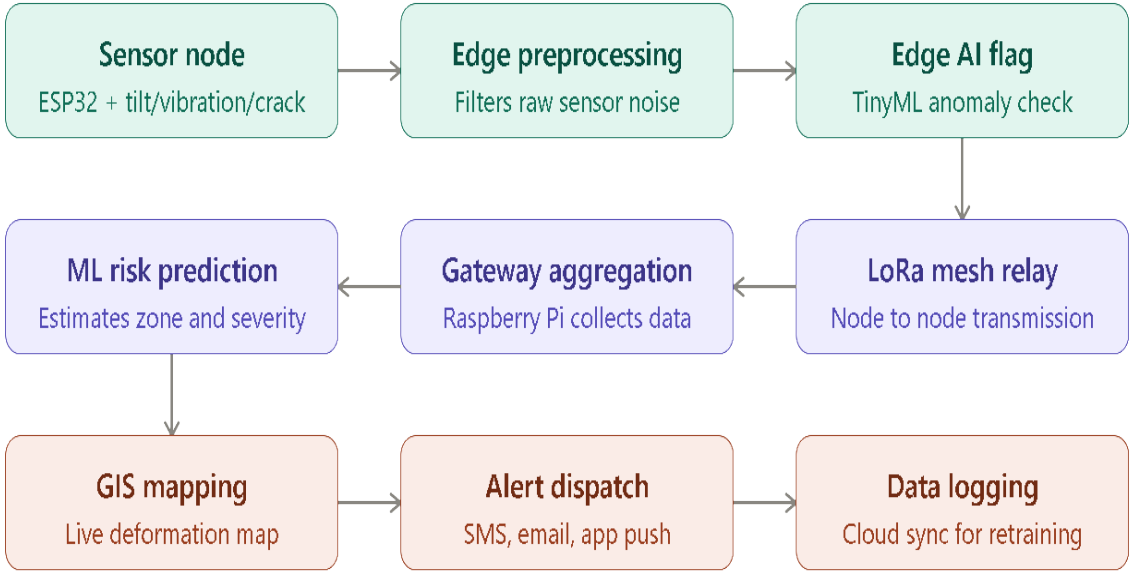
SMART INDIA HACKATHON 2026

Sensor data acquisition — tilt, vibration, displacement, crack

Edge preprocessing & filtering (noise reduction)

LoRa mesh transmission to gateway AI/ ML anomaly detection & severity prediction

GIS risk-zone mapping & alert generation



Implementation prototype paragraph:

"Our subsidence monitoring prototype is a self-contained mesh sensor network that captures tilt, vibration, displacement, and crack data entirely at the edge. AI models running on ESP32 nodes and a Raspberry Pi gateway detect abnormal deformation patterns and predict risk zones without needing constant internet access. A GIS dashboard and automated multi-channel alerts complete the real-time early warning loop.

FOR REFERENCE :
https://drive.google.com/file/d/1au_57OYMbq6X92KDbejyj3hBrqHyFe60/view?usp=drivesdk

Tich Titans

FEASIBILITY AND VIABILITY

SMART INDIA HACKATHON 2026

Harsh field environment: Dust, moisture, and temperature extremes affecting sensor durability on mine surfaces.

Mesh signal reliability: Terrain and vegetation causing LoRa signal attenuation or dropped nodes.

Power continuity: Sustaining always-on nodes in remote, often unelectrified deployment areas.

Model accuracy with limited data: Few labeled real-world subsidence events to train ML models initially.



Mitigation:

Ruggedized, IP65+ weatherproof node enclosures built for field deployment.
Redundant mesh topology so data reroutes automatically if a node fails or signal drops.
Solar panel + battery backup per node for uninterrupted low-power operation.
Start with simulated/synthetic subsidence data and transfer learning, refining continuously as real field data accumula

Tich Titans

IMPACT AND BENEFITS

SMART INDIA HACKATHON 2026

Impacts:

Real-time surface deformation monitoring gives mine operators and DGMS early warning windows before ground failure, protecting workers, communities, and infrastructure above mine panels.

Continuous mesh sensing replaces slow periodic surveys, eliminating blind spots between manual inspection cycles.

Benefits:

Social: Prevents loss of life and property from sudden collapse; builds public trust in mining safety.

Economic: Reduces damage-claim and rehabilitation costs; low-cost hardware makes it viable even for smaller mine operators.

Environmental: Limits uncontrolled land degradation and protects agricultural/ forest land above mine panels.

Quantifiable Analysis:

Enables detection of abnormal ground movement well before visible surface cracking, shrinking the response window from days/weeks (manual surveys) to real time.

Solar-powered nodes and LoRa's long range cut deployment/maintenance cost versus wired or satellite-based monitoring systems.

Tich Titans

RESEARCH AND REFERENCES

SMART INDIA HACKATHON 2026

Research papers on **mine subsidence monitoring and prediction**

Studies on **IoT-based underground mine monitoring**

Research on **AI/ML for anomaly detection and prediction**

Documentation on **ESP32 and sensor-based systems**

Research on **LoRa/LoRaWAN wireless communication**

References on **Edge AI / TinyML for real-time monitoring**

DGMS guidelines and mine safety standards

GIS-based methods for **risk-zone mapping and visualization**

IoT-based mine monitoring and sensor networks

AI/ML-based anomaly detection methods

Ground deformation and surface movement analysis

ESP32-based low-power sensor nodes

LoRa/LoRa mesh communication for long-range monitoring